Extending Unsupervised Landmark Discovery to Multi-Viewpoint Objects

Vaibhav Parekh (Advisor: Prof. Kenji Shimada)
Carnegie Mellon University

Carnegie Mellon University

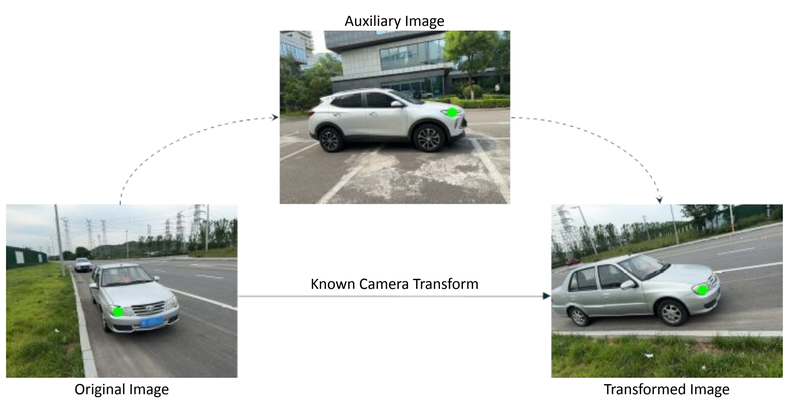
.CERLAB.

Introduction

We learn dense, view-invariant descriptors without any keypoint annotations, extending unsupervised landmark discovery from 2D faces to multi-viewpoint cars.

- Landmark discovery is essential for downstream perception tasks like pose estimation, but **manual annotation is expensive.**
- Most research revolves around human face/body datasets, **limiting applicability to single, near-front-view images**.
- *Thewlis et al., ICCV 2019* propose learning unsupervised landmarks on faces by **DVE (Descriptor Vector Exchange)** utilizing synthetic warps.
- **Real objects have full 360° viewpoint variation**, self-occlusion, and mirror symmetry, which synthetic TPS warps cannot capture.
- We **extend DVE to real multi-view** car data using LiDAR depth + camera geometry for correspondence and introducing visibility-aware loss normalization to handle occlusion.

DVE Transitive Matching



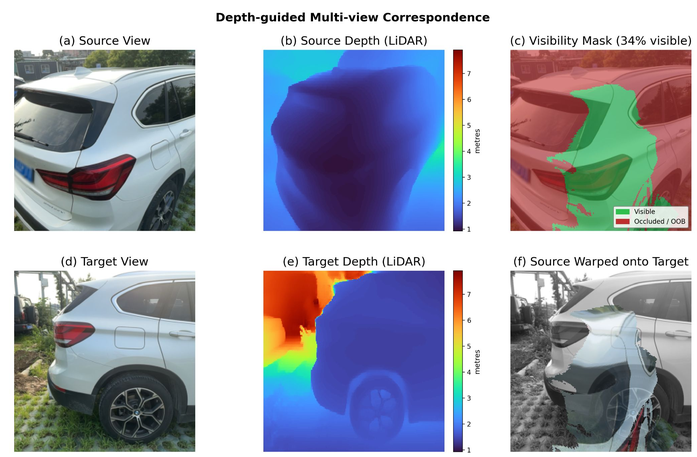
*The DVE mechanism **forces descriptors to generalize across car instances** by routing matching through an auxiliary image from a different car model .*

Best of Both Worlds

	Dimensions	View-invariant	Instance-invariant
Invariant Descriptors	C	X	
Dense Landmarks	2	X	X
DVE	C	X	X

Methodology

- A stacked Hourglass CNN maps each pixel to a point in 64-D embedding space, **where semantically corresponding landmarks produce similar vectors**.



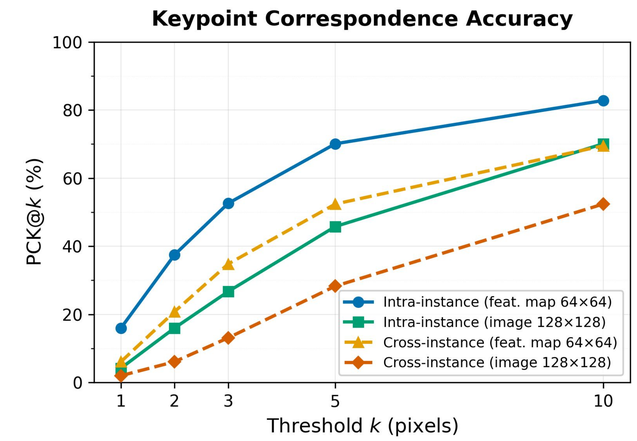
- The standard DVE loss normalizes over all H×W pixels, assuming every pixel to have a valid correspondence.
- But in real multi-view pairs, self-occluded and out-of-bounds pixels have no valid match.
- A **visibility-aware loss prevents noise from geometrically invalid correspondences** by normalizing only over visible pixels.

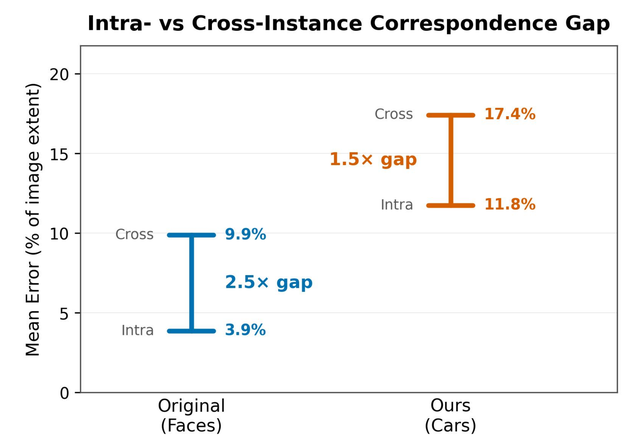
Qualitative Results

Intra-instance Keypoint Correspondence

Cross-instance Keypoint Correspondence

Quantitative Results





Conclusion

- **DVE successfully transfers from simpler facial data to real multi-view car data**, with PCK@10 of 82.8% for intra- and 69.5% for cross-instance correspondence.
- The intra- to cross-instance ratio of 1.5× suggests the DVE mechanism **generalizes effectively across car instances despite the harder domain**.
- At inference, the trained descriptors produce view-invariant features without any depth/multi-view input, **requiring only a single RGB image**.
- Future direction: replacing the CNN backbone with a contemporary transformer-based architecture.

Acknowledgement

- Dataset: 3DRealCar: An In-the-wild RGB-D Car Dataset with 360-degree Views
- Original Work: Thewlis et al. (ICCV 2019)